AARUSH DHINGRA
aarushdhingra27@gmail.com | +91 7088462345 | LinkedIn | GitHub

Profile Summary

Computer Science undergraduate specializing in **full-stack development and scalable systems**, with hands-on experience building applications using modern technologies. Skilled in **LLM integration, prompt engineering, and API development**, with a focus on performance, modular design, and usability. Passionate about building efficient, high-impact software solutions.

Education

Bachelor of Technology – Computer Science & Data Science (2023 – 2027)
CMR Institute of Technology, Bengaluru — CGPA: 8.71/10 (pursuing)

Technical Skills

- **Languages:** Python, Java, JavaScript, C
- **Frontend:** React.js, Next.js, Tailwind CSS, Electron, HTML, CSS
- **Backend:** FastAPI, Node.js, Express.js, RESTful APIs, WebSockets, JWT, OAuth 2.0
- **Databases:** PostgreSQL, pgvector, MySQL, MongoDB, Redis, Supabase, SQLite
- **Tools:** Docker, Git, GitHub, Postman, PowerShell, Vercel
- **AI/LLM:** RAG, LangChain, LangGraph, LLM Agents, Multi-Agent Systems, Tool Calling, MCP, Prompt Engineering, Embeddings, Vector Search, LLM Evaluation
- **Core Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Internship Experience

ThinkSeed Systems – Software Development Engineer Intern July 2026 – January 2027 (ongoing)

- Developing a **test automation framework** and software-based phone simulator to automate hardware-dependent validation workflows, reducing manual testing effort by **60%**.
- Automating simulations of **real-device interactions and communication flows**, cutting test execution time by **40%**.

Projects

Prmpt — AI Prompt Optimizer with Code RAG GitHub | Live *Python, FastAPI, Electron, SQLite FTS5*

- Built the hybrid retrieval engine for a code-aware RAG service (FastAPI), fusing dense embeddings, BM25 keyword search (SQLite FTS5) and concept search via Reciprocal Rank Fusion, with cross-encoder reranking and MMR.
- Designed a Graph RAG layer that models the codebase as a call/import graph, using PageRank to rank definitions above usages and Leiden community detection to answer repo-wide questions from cluster summaries.
- Built an evaluation suite (Recall@K, MRR, IoU, p50/p95 latency) on labelled queries and used it to make agentic multi-hop retrieval the default, raising **recall@5 from 0.32 to 0.80** and cutting median latency from **38s to 14s**.
- Built exact-function code search with **10/10 top-1 accuracy** at ∼280 ms, exposed as reusable REST APIs.

Hana Sensei - *AI Voice Agent for Learning Japanese* GitHub *Python, FastAPI, WebSockets, Whisper*

- Built a real-time voice agent that plans 5-day courses with an LLM, teaches through live speech, grades spoken answers and resumes from checkpoints after interruptions, at **∼1.0s voice-to-voice latency**.
- Designed a tiered grading pipeline (rules, Whisper forced alignment, LLM judge) with 12/12 correct accepts and 0 false accepts; a HuBERT phoneme model lifted mispronunciation detection from 2/13 to 8/13.
- Engineered guardrails and reliability: schema-validated outputs with retries, hedged requests with provider failover, deterministic fallbacks and time-to-first-token cut from 3–5s to ∼1s; 130+ pytest tests.

NetShape - *Network Condition Simulator* GitHub | Live *Python, asyncio, Typer, JavaScript*

- Built and published an open-source CLI tool to PyPI that wraps any process in a local HTTP/HTTPS forward proxy, injecting bandwidth throttling, latency, packet loss, and jitter with zero code changes to the target app.
- Implemented a token-bucket bandwidth throttling engine in Python `asyncio` with 12 built-in network profiles (2G through fiber) and real-time live adjustment without process restart.
- Added a live SSE dashboard, per-endpoint throttle rules and YAML scenarios for automated multi-phase runs.

Achievements

- Published **NetShape**, an open-source network simulation tool on PyPI.
- Ranked **Top 5** in BytesBattle Hackathon (CMRIT)
- Secured **Top 10** in TechAccelerate Hackathon (BITS Hyderabad)
- Launched **Prmpt**, a prompt optimization SaaS product.
- Solved **200+ DSA problems** on LeetCode